ROADWAY INFRASTRASTRUCTURE ECONOMIC RESILIENCE

EMGT 6930 FINAL PROJECT REPORT

Instructor: Yesim Sireli
Team 2: Harshavardhan Tumma, Kelly Sheffield, Sai Gnathavi Thadigoppula, Sierra Valder, Owen Zivitz

1.0 Abstract

This study evaluates the life cycle costs of asphalt (HMA) and concrete (PCC) pavement alternatives for a representative roadway segment in Charlotte, North Carolina, using real data from the North Carolina Department of Transportation (NCDOT). A 35 year analysis period is applied to quantify initial construction costs, maintenance, rehabilitation, and residual value. Net Present Value (NPV) is used to compare alternatives under varying economic and operational scenarios, including sensitivity to material costs and maintenance timing. Costs and impacts are assessed using guidance from the U.S. Department of Transportation. The findings demonstrate how life cycle cost analysis can support more informed and economically efficient pavement selection decisions for municipal transportation systems. Overall the findings showed that HMA has a lower initial and lifetime construction cost than PCC. Further analysis demonstrates benefits for PCC when external system interactions are included in the life cycle analysis.

Table of Contents

1.0 Abstract (pg. 1)
2.0 Background (pgs. 2 - 3)
2.1 Pavement Alternatives
2.2 Pavement Lifecycles
2.3 Charlotte Roadway Infrastructure
3.0 Elements of the Problem (pgs. 3 - 7)
3.1 Problem Statement
3.2 Objectives
3.3 Methodology
3.3.1 Analysis Methods
3.3.2 Study Period
3.3.3 Unequal Lives of Alternatives
3.4 Lifecycle Data Inputs
3.4.1 MARR
3.4.2 Initial Cost, O&M, and Residual Value
3.4.3 MARR
3.4.4 Sustainability
4.0 Results and Discussion (pgs. 7 - 13)
4.1 NPV and Annual Worth
4.2 Societal Benefits
4.3 Sustainability
4.4 Usage-Based Rehabilitation Scenario
4.5 Oil Market Cost Scenario
4.6 Case Study Application - Bryant Farms Road Extension
4.7 AI Application Discussion
4.8 Sensitivity Analysis
5.0 Conclusion (pgs. 13 - 14)
6.0 References (pgs. 14 - 15)

2.0 Background

2.1 Pavement Alternatives

Hot mix asphalt (HMA) and Portland cement concrete (PCC) are two of the most used pavement materials for roads in North Carolinas. PCC is a rigid cement pavement that is known for its durability and long lifetime. HMA is a flexible pavement composed of bituminous binder that is the most common option in the state due to its lower upfront cost and historical preference (FHWA, 1998).

2.2 Pavement Lifecycles

The lifecycles of pavements follow a pattern of initial construction, minor repairs, major repairs, and the eventual retirement of the roadway surface. This pattern does change depending on the pavement surface that is selected. Asphalt pavement, due to its soft and flexible nature, requires frequent rehabilitation. Concrete pavement does not need this major repair during its lifecycle, instead the surface can be refreshed with diamond grinding. At the end of both pavements' useful life the surface and subbase will be fully removed and reconstructed (Robbins & Tran, 2018).

2.3 Charlotte Roadway Infrastructure

The City of Charlotte has over 5,500 lane miles under their jurisdiction. They are responsible for both capital and O&M funding for these municipal roads. Charlotte's 2026 fiscal budget includes a total of $55.1 million in allocated capital funding and $37.6 million for operations with an additional $15 million for street maintenance alone (City of Charlotte, 2025).

3.0 Elements of the Problem

3.1 Problem Statement

Current pavement selection decisions fail to evaluate pavement alternatives as a long-term system with external interactions. The investment decision has financial impacts that compound over decades through maintenance cycles, market exposure, and infrastructure disruptions. This project aims to improve the economic resilience of the City of Charlotte's roadway infrastructure.

3.2 Objectives

The purpose of this project is to compare pavement alternatives using life-cycle cost, scenario analysis and sensitivity analysis. The specific objectives are as follows:

1. Quantify the pavement life cycle costs including initial costs, maintenance, major repairs and residual value.
2. Apply capital costing methods, including NPV, to support comparison of pavement alternatives under different economic and operational scenarios.
3. Evaluate uncertainty and decision robustness using scenario-based and sensitivity analysis.
4. Quantify societal and sustainability impacts associated with pavement alternatives.
5. Provide recommendations for Charlotte infrastructure decisions on future pavement selection and financial planning.
6. Apply the analysis to a representative Charlotte roadway case study to compare direct agency costs with broader societal and environmental impacts.

3.3 Methodology

3.3.1 Analysis Methods

The focus of this project was to determine the lifecycle costs of pavement alternatives and analyze the variability of this cost through different scenarios. This is the basis for efficient and cost-effective pavement selection. Concepts learned in this class were utilized to understand each of these alternatives economically. These concepts are Net Present Value (NPV) Analysis, Scenario Analysis, Societal Benefit, and Sustainability Analysis (Rafael Bañares-Alcántara et al. 2021).

3.3.2 Study Period

The study period is based on a 35-year lifecycle analysis. This is the minimum amount of time recommended by the Federal Highway Administration for conducting roadway studies. This study period is near the useful life of pavement surfaces of 30-50 years and allows for enough time for long-term interactions to take place. The estimated design life of HMA asphalt was found to be around 35 years depending on when the final rehabilitation took place. For PCC the average design life was found to be 40 years for the purposes of this study, although some studies have found PCC lasting over 50 years (FHWA, 1998).

3.3.3 Unequal Lives of Alternatives

HMA and PCC have unequal lives which need to be leveled to have accurate analysis. PCC pavement is estimated to have a useful life of around 40 years (Robbins & Tran, 2018) while asphalt pavement has a lifespan of around 30 years. The lifespan of these two alternatives can also vary based on maintenance plans and designs. This particularly influences asphalt's lifespan due to its reliance on rehabilitation activities. To balance these two uncertain lives, an approach was used which determines the residual value shown in figure 2.

Figure 1. Residual Value Formula

$$Residual\ Value = \frac{Expected\ Life\ of\ Rehab - Time\ Since\ Rehab}{Expected\ Life\ of\ Rehab} \times Cost\ of\ Rehab$$

3.4 Lifecycle Data Inputs

3.4.1 MARR Inputs

The MARR of 4.1% is sourced from the 30-year nominal interest rate on United States Treasury Notes as published by the Office of Management and Budget (OMB, 2026).

This MARR was chosen as it will represent a long-term benchmark for relatively low-risk public investments.

3.4.2 Initial Cost, Maintenance, and Residual Value Inputs

All cost inputs are estimated from North Carolina Department of Transportation (NCDOT) bid data (NCDOT, n.d.). Three competitive bid events per material type, from July 2023 through September 2026, were averaged per event and adjusted to present value a base year using the 4.1% MARR. Unit prices in dollars per ton were converted to dollars per square yard using cross-section depths and standard material density factors per the NCDOT Design Manual. These inputs are shown below in Table 1.

Table 1 LCCA Cost Inputs by Pavement Type (NCDOT Bid Tab Data)

Cost Component	HMA ($/SY)	PCC ($/SY)	Frequency
Surface Course	$35.87	$103.13	Initial Cost
Aggregate Base Course (ABC)	$26.60	$19.95	Initial Cost
Total Initial Cost	**$62.46**	**$123.08**	Initial Cost
Mill and Overlay / Diamond Grinding (per event)	$35.36	$3.86	12 / 10 years
Crack Sealing (HMA) / Joint Sealing (PCC)	$0.70	$0.20	2 / 5 years
Residual Value at Year 35	$17.68	$15.38	End of Study

3.4.3 Carbon Embodiment Inputs

Carbon embodiment was included to estimate the emissions directly associated with the pavement materials used in initial construction and rehabilitation. Emission factors were applied in lb CO_2 per ton of material and converted to lb CO_2 per square yard using the pavement section thickness and material density utilized in the NPV analysis (City of Portland, 2022), (Gruber & Hofko, 2023). Utilizing the environmental cost of carbon published by the EPA (IWGSCGG, 2016) , an estimated cost can be associated with the embodied carbon. For this project 100$/Ton was utilized. These estimations are shown below in table 2.

Table 2 Material Carbon Embodiment

Input	HMA	PCC
Pavement section	5.5 in	6.0 in
Material quantity	0.30 tons/SY	0.34 tons/SY
CO_2 emission factor	445 lb CO_2/ton	270 lb CO_2/ton
Calculated Lifecycle embodiment	280 lb CO_2/SY	91 lb CO_2/SY
Carbon cost factor	\$100/ton CO_2	\$100/ton CO_2
Calculated carbon cost PV	\$14.00/SY	\$4.55/SY

4.0 Results and Discussion

4.1 NPV and Annual Worth Analysis

The NPV analysis was used to compare the total lifecycle costs of PCC and HMA. The 35-year lifecycle analysis found HMA to have a total cost of \$110.46 \$/SY and PCC to be \$125.04 \$/SY. HMA also has a \$60.62/SY lower initial construction cost than PCC. However, each HMA mill-and-overlay rehabilitation event costs \$35.36/SY compared to \$3.86/SY for PCC diamond grinding. This is a large cost differential which can accumulate over time. The lifecycle costs were also annualized over the 35-year lifecycle at the 4.1% MARR. This found an annual payment of \$6.79 Per SY for PCC and \$6.00 Per SY for HMA. This cost represents the yearly cost to the municipality each year during its lifecycle. The PCC under standard conditions is estimated to cost \$0.79 per SY per year over HMA

4.2 Societal Benefits

Pavement selection influences not only costs but also broader societal outcomes, particularly through user delay during maintenance activities. Work zones reduce roadway capacity and increase travel time, creating costs for the public. Following

guidance from the United States Department of Transportation, user delay costs can be estimated by multiplying total delay hours by the value of time (U.S. Department of Transportation, 2025). For a representative segment of Bryant Farms Road, this study assumes an Average Daily Traffic (ADT) of 20,000 vehicles, a 15-minute delay per vehicle, and a value of time of $21.10 per hour. Asphalt overlays are assumed to require 5 days of work, while concrete repairs require 2 days. These assumptions produce the per-event user costs shown in Table 3.

Table 3 Societal Cost

Pavement Type	Duration (Days)	Unit Cost Per Event ($)	Maintenance Years
Asphalt (HMA)	5	527,500	12, 22, 32
Concrete (PCC)	2	211,000	10, 20, 30

Over the 35-year analysis period, asphalt results in higher cumulative user costs due to the longer duration maintenance events. In contrast, concrete has shorter disruptions, resulting in lower total user delay costs. This highlights a key tradeoff between maintenance frequency and disruption intensity. In addition to delay, pavement selection can influence fuel consumption, emissions, and roadway safety, as work zones are associated with increased crash risk (Federal Highway Administration, 1998). While these impacts are not quantified in this study, they further support the importance of incorporating societal considerations into pavement investment decisions.

4.3 Sustainability

Sustainability was evaluated using a lifecycle embodied CO_2 to quantify the lifecycle environmental cost.The analysis used lbs of CO_2 embodied from material per SY of surface. Based on the lifecycle modeling, HMA generates approximately 275 lbs CO_2 per

square yard over 35 years, while PCC generates approximately 100 lbs CO_2 per square yard (City of Portland, 2022), (Gruber & Hofko, 2023). The difference is primarily driven by the frequency of resurfacing cycles in HMA, which significantly increases material usage and associated emissions. To interpret these results in economic terms, embodied CO_2 values were converted to an environmental cost using a carbon price assumption of $100 per ton of CO_2. This results in an estimated environmental cost of approximately $14 per square yard for HMA and $5 per square yard for PCC. The results demonstrate that maintenance-related material demand is the dominant factor influencing embodied carbon over the lifecycle. Pavement alternatives with frequent resurfacing cycles accumulate significantly higher environmental impact.

4.4 Usage-Based Rehabilitation Scenario

Pavement surfaces have an uncertain long-term truck loading which can greatly influence useful life and rehabilitation cycles. Three probabilistic scenarios were defined using pavement performance data and published LCCA research. Rehabilitation intervals are derived from NCDOT performance-based pavement management data and are consistent with service life findings reported in the literature for HMA and jointed plain concrete pavement (JPCP) in North Carolina (Robbins, 2018; Sheffield, 2025). Under high use, HMA requires resurfacing every 8 years, producing four mill-and-overlay events over 35 years. PCC requires diamond grinding at 18-year intervals, producing one event. Under average use, three HMA events and one PCC event occur. Under low use, two HMA events and one PCC event occur. Table 2 shows the intervals and Table 3 presents the NPV results.

Table 2. Rehabilitation Intervals and Event Counts by Usage Scenario

Scenario	Probability	HMA Interval (yrs)	PCC Interval (yrs)	HMA Events / 35 yrs
High Use	25%	8	18	4
Average Use	50%	11	24	3
Low Use	25%	14	30	2
Weighted Average	100%	11	24	--

Table 3. NPV Results by Usage Scenario (35-Year Analysis, $/SY)

Scenario	HMA NPV ($/SY)	PCC NPV ($/SY)	Difference (PCC - HMA)	Lower Cost
High Use	137.49	125.62	-11.87	PCC
Average Use	116.48	125	9	HMA
Low Use	101.11	125	24	HMA
Prob.-Weighted Expected	117.89	125	7	HMA

The probability-weighted expected NPV favors HMA by $7.35/SY, anchored by the 50% weight on the average use scenario. The critical risk is a high-use outcome: if road use exceeds projections, PCC becomes the lower-cost alternative by $11.87/SY.

4.5 Oil Market Cost Scenario

Asphalt binder is a petroleum refining byproduct, meaning every future HMA resurfacing event carries oil price risk that PCC does not share. Asphalt prices have experienced fluctuations exceeding 40% annually over the past 50 years, reinforcing the need to evaluate petroleum market volatility as a distinct risk factor in HMA lifecycle cost analysis (Sheffield, 2025). Historical crude oil price data from the U.S. Energy Information Administration confirms sustained periods of both price elevation and depression that materially affect asphalt binder costs (U.S. Energy Information

Administration, 2026) . Three oil market conditions were modeled by applying a cost multiplier to all HMA costs throughout the analysis period. PCC costs are unaffected in all scenarios. Table 4 presents the parameters and NPV results.

Table 4. Oil Market Scenario Parameters and NPV Results (Average Use Intervals, $/SY)

Oil Market Condition	Probability	HMA Multiplier	HMA NPV ($/SY)	PCC NPV ($/SY)	Lower Cost
High Cost (+100% of avg)	25%	1.70x	$193.52	$125.22	PCC
Average Cost (Base)	50%	1.00x	$116.48	$125.22	HMA
Low Cost (-50% of avg)	25%	0.50x	$61.45	$125.22	HMA
Prob.-Weighted Expected	100%	1.05x	$121.98	$125.22	HMA

PCC NPV holds constant at $125.22/SY across all oil conditions because its cost structure contains no petroleum component. HMA NPV swings from $61.45/SY to $193.52/SY, a range of $132.07/SY driven entirely by oil market conditions. The probability-weighted expected NPV favors HMA by $3.24/SY, though a sustained high-oil-cost environment reverses the decision in favor of PCC by $68.30/SY.

4.6 Case Study Application - Bryant Farms Road Extension

The Bryant Farms Road Extension was used to apply the lifecycle cost, societal cost, and sustainability results to represent a Charlotte roadway project. The project was modeled as a 1.2-mile roadway with a total pavement area of 25,344 SY. The expanded NPV analysis includes direct NPV, embodied carbon cost, and user delay cost. This allows the pavement alternatives to be compared based on the total lifecycle cost experienced by both the municipality and the public. This combined NPV found a total project cost for PCC to be $3.59 million and HMA to be $3.84 million. This is a lifetime cost savings of nearly $250,000 over the project lifetime. The full cost breakouts can be seen in table 5.

Table 5. Bryant Farms Road Expanded NPV Results

Cost Category	HMA ($/SY)	PCC ($/SY)	HMA Total Cost	PCC Total Cost
Agency NPV	110.46	125.04	2,799,498	3,168,014
Embodied carbon cost	14	5	348,480	126,720
User delay cost	27	12	689,357	298,794
Expanded NPV	**151**	**142**	**3,837,335**	**3,593,528**

When the expanded NPV is applied to the scenario analysis, PCC is already a lower cost option over the Bryant Farms Road Extension project. The scenario analysis further shows PCC's cost benefit under uncertain future conditions. With the road usage scenario analysis, the PCC has an estimated cost reduction of $21.45/SY or around $544,000 across the project lifecycle. Under the high-oil-cost scenario, the additional PCC cost benefit of $68.30/SY increases the total difference to $77.88/SY, or approximately $1.97 million across the project. This shows that PCC not only is the lower expected combined NPV, but it also provides greater resilience against poor economic and traffic load conditions.

4.7 AI Application Discussion

In this study Artificial intelligence improves the accuracy of the analysis inputs through mass data aggregation and deterministic analysis. Pavement construction and rehabilitation cost estimation is supported by historical NCDOT bid tabulation data. Analysis of this data identifies trends and variability in material prices, particularly for asphalt pavements, which are influenced by fluctuations in petroleum markets. Data-driven models provide improved estimates of future costs used in lifecycle analysis. Pavement performance databases also support the modeling of deterioration and rehabilitation needs. Overall, AI functions as an analytical support tool that improves

input estimation, captures system variability, and strengthens decision-making in pavement lifecycle analysis (Bañares-Alcántara et al., 2021).

4.8 Sensitivity Analysis Discussion

Sensitivity analysis was used to test how changes in key assumptions affect the pavement decision. The base case favors HMA, with a present value cost of $110.46/SY versus $125.04/SY for PCC, giving HMA a savings of $14.59/SY. One-at-a-time analysis showed that PCC initial cost, HMA initial cost, HMA maintenance present value, and MARR are the most influential decision drivers, while PCC maintenance and salvage value have little effect. Two-at-a-time analysis further showed that when HMA initial cost and HMA maintenance cost rise together, the HMA advantage can shrink and reverse. Finally, optimistic–most likely–pessimistic analysis confirmed that HMA remains preferred under expected conditions, while PCC becomes more attractive only under unfavorable combined HMA cost conditions. These results show that sensitivity analysis supports not only the base-case comparison, but also the robustness of the final pavement recommendation.

5.0 Conclusion

This study evaluated HMA and PCC pavement alternatives using NPV, scenario analysis, societal impacts, sustainability, and sensitivity analysis. Under direct agency-cost analysis, HMA is the lower-cost alternative, with an NPV of $110.46/SY compared to $125.04/SY for PCC, and it remains the expected lower-cost option in both the usage-based and oil-market scenarios. However, when user delay and embodied carbon costs are included in the Bryant Farms Road Extension case study, PCC becomes the lower total-cost alternative, with an expanded NPV of $3.59 million compared to $3.84 million

for HMA. Sensitivity analysis shows that the decision is mainly driven by initial costs, HMA maintenance cost, and MARR, and that the HMA advantage can reverse under unfavorable combined cost conditions. Overall, PCC provides greater long-term resilience when broader economic, societal, and environmental impacts are considered.

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